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Assignment-1

1

1. The role of yield is the representation of good dies from the wafer.

For example: if the yield is 80%, then 80 out of 100 dies from the wafer is good.

2. wafer area $= \pi r^2$

for wafer A,

$d = 30\,cm$

$\therefore r = 15\,cm$

no. of dies = 150

$\therefore$ die area $= \frac{\pi \times (15)^2}{150}$

$= 4.71\,cm^2$

for wafer B,

$d = 25\,cm$

$r = 12.5\,cm$

no. of dies = 120

$\therefore$ die area $= \frac{\pi \times (12.5)^2}{120}$

$= 4.09\,cm^2$

3. $\text{yield}_A = \frac{1}{\left(1 + \left(\frac{0.015 \times 4.71}{2}\right)\right)^2}$

$= 0.933 \times 100$

$= 93.3\%$

$\text{yield}_B = \frac{1}{\left(1 + \left(\frac{0.02 \times 4.09}{2}\right)\right)^2}$

$= 0.923 \times 100$

$= 92.3\%$

for A,
defect per area = 0.015 cm

for B,
defect per area = 0.020

4: Cost per die = $\dfrac{\text{Cost per wafer}}{\text{Dies per wafer} \times \text{Yield}}$

given
Cost for wafer$_A$ = 30 \$
cost for wafer B = ~~25~~ 25 \$

for Wafer A = $\dfrac{30}{(150 \times 0.933)}$ = 0.214 \$

for, wafer B = $\dfrac{25}{(120 \times 0.923)}$ = 0.226 \$

5.
Wafer A is more cost effective as
0.219 \$ < 0.226 \$

2.

1: Average CPI
for snapX = $(0.4 \times 2) + (0.3 \times 3) + (0.2 \times 1) + (0.1 \times 5)$
= 2.4

for mediaz = $(0.4 \times 1) + (0.3 \times 4) + (0.2 \times 2) + (0.1 \times 2)$
= 2.2 (Am)

2: Execution time
$T = \dfrac{IC \times CPI}{\text{clockrate}}$

CamScanner

for Snap X,

$$T = \frac{IC \times CPI}{clockrate}$$

$$= \frac{2\times10^6 \times 2.4}{2.2\times10^9}$$

$$= 2.18 \times 10^{-3} s$$

$$= 2.18 \text{ ms}$$

given,
clockrate = 2.2 GHz
IC = 2 million
= 2×10^6
CPI = 2.4 [from 1]

for media z,

$$T = \frac{IC \times CPI}{clockrate}$$

$$= \frac{2\times10^6 \times 2.2}{2.5\times10^9}$$

$$= 1.76 \text{ ms}$$

given,
clockrate = 2.5 gHz
= 2.5×10^9
IC = 2 m = 2×10^6
CPI = 2.2

3.
~~form~~ from '2' we see, Snap X takes more time to complete the task.

∴ more time taken = $(2.18 - 1.76)$ ms
= 0.42 ms

∴ media z has better perf ~~by~~

$$\text{Percentage} = \left(\frac{2.18 - 1.76}{2.18}\right) \times 100 = 19.3\%$$

∴ media z has better perf by 19.3%.

4

for snapX,

$$SPEC = \frac{100}{2.18} = 95.87$$

Reference time = 100ms

for mediaZ,

$$SPEC = \frac{100}{1.76} = 56.82$$

3.

1. $Q1 = 2.8\ gHz$

$CPIQ_1 = 1.8$

To improve performance by 40%, new execution time $T_2 = \frac{T_1}{1.4}$

$$\text{Ratio} = \frac{T_{old}}{T_{new}} = 1.4$$

2. Required clock rate to meet ~~goal~~ goal = f_2

Now,

$$T = \frac{IC \times CPI}{f}$$

for,

$$T_{old} = \frac{IC \times 1.8}{2.8}$$

$$T_{new} = \frac{IC \times \cancel{1.8}\ 2.2}{\cancel{2.8}\ f} \times 1.4$$

Perf f_2 = Perf $f_1 \times 1.4$

$$\frac{f_2}{2.2} = 1.4 \times \left(\frac{1.8}{2.8}\right)$$ [considering same ISA)

$$\frac{f_2}{2.2} = 1.4 \times 1.556$$

B

$$\Rightarrow \frac{f_2}{2.2} = 2.178$$

$$\Rightarrow f_2 = 4.79 \text{ GHz}$$

3.

The tradeoff is justified if certain performance goals are met. Otherwise, the higher clock rate will generate more heat and require ~~more~~ better cooling solutions.

4.

P = capacitive lad × voltage^2 × frequency

for v_1,

$$P = 100 \times 1.2^2 \times 2.5 = 360 \text{ W}$$

for v_2,

$$P = 80 \times 0.85^2 \times 2.2 = 127.16 \text{ W}$$

$$\text{Reduction} = \left(\frac{360 - 127.16}{360}\right) \times 100 = 64.68\%$$

If the original system used 100W, the new system will use,

$$\cancel{100 \times \left(\frac{360}{127.16}\right)} \quad 100 \times \left(\frac{127.16}{360}\right) = 35.3\text{W}$$

5.

1) Sys A, $T = \frac{IC \times CPI}{\text{clockrate}}$

$= \frac{\cancel{9}^{2}\times\cancel{10^9}\times\cancel{1.5}^{1}}{\cancel{3\times10^9}_{\cancel{2}\,1}}$

$= 2$ sec

Sys A,
$IC = 9b = 9\times10^9$
$CPI = 1.5$
Clock rate $= 3$ gHz
$= 3\times10^9$

Sys B, $T = \frac{IC \times CPI}{\text{clockrate}}$

$= \frac{1.5\times\cancel{10^9}\times 2}{2.5\times\cancel{10^9}}$

$= \frac{3}{2.5} = 1.2$ sec

Sys B,
$IC = 1.5b = 1.5\times10^9$
$CPI = 2.0$
clockrate $= 2.5\times10^9$

2) MIPS for sys A $= \frac{3\times10^9 \; \cancel{3\,GHz}}{1.5\times10^6} = 2000$ MIPS

for sys B $= \frac{\cancel{3G\ 2\times10^9}}{\cancel{2}} = \frac{2.5\times10^9}{2\times10^6} = 1250$ MIPS

3 Execution time for sys A = 2 sec

" " " " B = 1.2 "

As $2 > 1.2$

∴ Sys B has better ~~for~~ performance

4

Higher MIPS does not always imply better performance because:

1. Ignores the IC
2. Ignores the complexity of instruction, workload size etc.
3. Ignores if there are different ISA.

6

1

$$\text{Total time} = (25 \times 300) + (15 \times 500)$$
$$= 7500 + 7500$$
$$= 15000 \text{ ps}$$
$$= 15 \text{ ns}$$

(Ans)

2 Clock period = 100 ps

$\therefore$ Type-X CPI $= \frac{300}{100} = 3$

$\therefore$ Type-Y CPI $= \frac{500}{100} = 5$

Total instructions
$= 25 + 15$
$= 40$

$$\text{Average CPI} = \frac{(25 \times 3) + (15 \times 5)}{40} = 3.75$$

(Ans)

3 SPEC ratio = $\frac{14.2 ns}{15 ns}$ = ~~0.947~~ 0.947

9

New spec ratio = 1.5

$$1.5 = \frac{14.2}{T_{new}}$$

$$\Rightarrow T_{new} = \frac{14.2}{1.5} = \cancel{96}9467 ps$$

Now,

$$9467 = (25 \times T_x) + (15 \times 500)$$

$$\Rightarrow 9467 = (25 \times T_x) + 7500$$

$$\Rightarrow 9467 = 25T_x + \cancel{7525}\ 7500$$

~~$\Rightarrow T_x = 1942$ ps~~

$$\Rightarrow 25T_x = 9467 - 7500$$

$$\Rightarrow T_x = \frac{9647 - 7500}{25} = 78.68 ps.$$

7.

1. Yes, the performance can be increased by 20% by only reducing the execution time fo SB type.

for 20% ~~impo~~ improvement,

total time = 180/1.2 = 150 sec

Remaining after R-type and I/S type instructions

= (150 − (80+45))

= 150 − 125 = 25 sec

∴ SB must be improved by = $\frac{55}{25}$ = 2.2 times

2.

IC − Most affected. Because, higher level and lower level languages generate different number of instruction. An optimized language/compiler reduces IC significantly.

CPI − The language decides which instruction types are used. So, it is also affected but not significantly.

Clock rate − Hardware property. Not affected.

3.

Performance via prediction is making an educated ~~based~~ guess based on the outcome of an instruction before it has run.

For example: in branching or if/else statements, the CPU guesses whether an if statement will be true/false to generate instructions.

4. Redundancy ensures system reliability and is necessary for a backup. For example: in windows/linux we can take backup of our whole system. So, if in the future if something goes wrong the system can be reverted to the backed up state